$\pi_4(S^3) = \mathbb{Z}/2$ THROUGH THE DELOOPING EMBEDDING: THE MAIN IDEAS

Attribution. The ingredients are David Wärn's: a candidate Postnikov stage BR identified by a *marked universal property*, and the fact that *delooping is an embedding* of pointed 2-connected 4-types into pointed simply connected 3-types. The proof below uses them, but it is *not* the route Wärn had in mind, which goes through a classification of arbitrary maps between triples; we bypass that step with specialised mapping arguments (Section 7). The delooping machinery (Eilenberg–MacLane spaces as iterated torsors, the criterion for a map of loop spaces to deloop) is Wärn's [1]. The classification of pointed simply connected 3-types by triples is Hart–Milner's (Owen Milner's equivalence) [2]; the development re-proves it in that case. The cup-square calculations used underneath (cross term, bilinear classification, Lemma N) are those of the earlier proof, `doc/pi4-s3.pdf`.

The result. In univalent type theory with replacement, assuming only the universal property of S^3,

$$\pi_4(S^3) \cong \mathbb{Z}/2 \quad \text{as groups}$$

(`Pi4Direct.Pi4S3.pi4-S3`). It is a second proof of the theorem of `doc/pi4-s3.pdf` (`Pi4Sphere.Pi4S3.pi4-S3`): the two share the candidate BR, the marked class s, and the final Yoneda argument, but differ in how the universal property at $\|S^3\|_4$ is obtained. The earlier proof builds a Postnikov presentation of $\|S^3\|_4$; the present one proves the universal property at *every* pointed 2-connected 4-type, from a restricted 2-torsion lemma, and needs no presentation.

1. Where the argument goes

Notation. $\mathrm{Map}_*(X,Y)$ is the type of pointed maps, $[X,Y]_*$ its set truncation, $c;d$ composition of classes. $K(A,n)$ are the Eilenberg–MacLane spaces, $e_m : \Omega K(A,m{+}1) \simeq_* K(A,m)$. $\kappa : K(\mathbb{Z},2) \to_* K(\mathbb{Z},4)$ is the constructed cup square, $\rho : \mathbb{Z} \to \mathbb{Z}/2$ the parity, and $\bar\kappa = \kappa; K(\rho,4)$. For a pointed map k, Fib$k$ is its fibre and $\mathsf{fpr} : \mathrm{Fib}k \to_*$ (source of k) the projection.

The candidate and the final step. Let $R = \mathrm{Fib}\bar\kappa$, a pointed simply connected 3-type with $\pi_3 R = \mathbb{Z}/2$. The cross term of $\bar\kappa$ vanishes, so $\bar\kappa$ deloops to $\hat F : K(\mathbb{Z},3) \to_* K(\mathbb{Z}/2,5)$ and $BR = \mathrm{Fib}\hat F$ is a pointed 2-connected 4-type with $\Omega BR \simeq_* R$; in particular $\pi_4 BR \cong \pi_3 R \cong \mathbb{Z}/2$. There is a class $s \in [S^3, BR]_*$ (`SourceS3.Final.s`). Write

$$\mathsf{M}(Y): \qquad s^* : [BR,Y]_* \to [S^3,Y]_*,\ c \mapsto s;c, \quad \text{is a bijection.}$$

If $\mathsf{M}(BR)$ and $\mathsf{M}(T)$ hold for $T = \|S^3\|_4$, a two-object Yoneda argument on classes gives inverse classes $T \leftrightarrows BR$, hence $\pi_4 S^3 \cong \pi_4 T \cong \pi_4 BR \cong \mathbb{Z}/2$ (`Pi4Sphere.Two4`). $\mathsf{M}(BR)$ is the mapping theorem for targets Fib$K$ of level-3 triples (`SourceS3.Final.rep-fib`). The whole difficulty is $\mathsf{M}(T)$.

The new route. We prove $\mathsf{M}(Y)$ for every pointed 2-connected 4-type Y:

(1) every such Y has $\Omega Y \simeq_* \mathrm{Fib}k$ for a *deloopable* triple (A,B,k) (Section 2);
(2) $\mathsf{M}(Y)$ is equivalent to its looped form, a statement about maps into ΩY (Section 3);
(3) the looped form reduces, by a lifting lemma along fpr, to one numerical fact about deloopable triples (Section 4);

Date: 28 October 2026.

(4) that fact is Statement T, the restricted 2-torsion lemma (Section 5): a cup-square calculation in which deloopability enters only through the multiplicativity of one loop map.

2. Delooping is an embedding

Let SC_4 be the type of pointed 2-connected 4-types and SC_3 that of pointed simply connected 3-types; looping is a map $\Omega : \mathsf{SC}_4 \to \mathsf{SC}_3$.

Lemma 1 (looping of maps)**.** *If X is 2-connected and Y is 4-truncated, then* $\mathsf{loop} : \mathrm{Map}_*(X, Y) \to \mathrm{Map}_*(\Omega X, \Omega Y)$ *is an equivalence.*

This is Wärn's delooping criterion [1]: the type of deloopings of a map of loop spaces is contractible under these connectivity and truncation bounds (`SourceS3.LoopEquiv`).

Theorem 2 (Wärn)**.** $\Omega : \mathsf{SC}_4 \to \mathsf{SC}_3$ *is an embedding: its fibres are propositions.*

By Lemma 1 and Whitehead, looping of pointed equivalences $(u \simeq_* v) \to (\Omega u \simeq_* \Omega v)$ has a section; so the type of v with $\Omega u \simeq_* \Omega v$ is a retract of the (contractible) type of v with $u \simeq_* v$, and the fundamental theorem of identity types gives the claim (`DeloopEmbedding.Emb`).

By the classification of Hart–Milner, $\mathsf{SC}_3 \simeq \mathsf{Triples}$, where a triple (A, B, k) consists of abelian groups A, B and $k : K(A, 2) \to_* K(B, 4)$, realised as $\mathrm{Fib}k$ (`PostnikovClassification.classification`). Call a triple *deloopable* if $\mathrm{Fib}k$ is in the image of Ω; by Theorem 2 this is a proposition, and every $Y \in \mathsf{SC}_4$ is a delooping of the triple classifying ΩY:

$$\textstyle\sum_{t:\mathsf{Triples}} \mathsf{Deloopable}(t) \;\simeq\; \mathsf{SC}_4.$$

So it suffices to understand maps into $\Omega Y \simeq_* \mathrm{Fib}k$ for deloopable triples. Note that being deloopable is weaker, a priori, than k itself being the looping of a map $K(A, 3) \to K(B, 5)$; we never need to decide this.

3. The looped form of M

Let $r = (\mathsf{loop}s); [\Omega BR \simeq_* R] \in [\Omega S^3, R]_*$, the looping of the marked class, viewed with target R (`MarkLoop.LMark.r`).

Proposition 3 (L)**.** *For $Y \in \mathsf{SC}_4$: $\mathsf{M}(Y)$ holds iff $r^* : [R, \Omega Y]_* \to [\Omega S^3, \Omega Y]_*$ is a bijection.*

Looping commutes with precomposition, and by Lemma 1 looping is a bijection $[BR, Y]_* \to [\Omega BR, \Omega Y]_*$ and $[S^3, Y]_* \to [\Omega S^3, \Omega Y]_*$ (BR and S^3 are 2-connected). Two applications of 2-out-of-3 conclude (`MarkLoop.LMark.At.to-loop`, `from-loop`).

The point of the looped form is that its target $\Omega Y \simeq_* \mathrm{Fib}k$ is a fibre, and maps into a fibre are analysed by a lifting lemma.

4. Reduction to a numerical fact

Fix a deloopable triple (A, B, k). Postcomposition with fpr gives, for every pointed source Q, a map $q_Q : [Q, \mathrm{Fib}k]_* \to [Q, K(A, 2)]_*$, and the square

$$q_{\Omega S^3}(r; c) = r; q_R(c)$$

commutes. We show that q_R, $q_{\Omega S^3}$ and $r^* : [R, K(A, 2)]_* \to [\Omega S^3, K(A, 2)]_*$ are bijections; then r^* is a bijection into $\mathrm{Fib}k$ by 2-out-of-3, and into ΩY by transport along $\Omega Y \simeq_* \mathrm{Fib}k$; Proposition L gives $\mathsf{M}(Y)$.

r^* into $K(A, 2)$. $K(A, 3) \in \mathsf{SC}_4$ and $\mathsf{M}(K(A, 3))$ holds (it is part of the construction of s: $s; [p] = [\sigma_3]$ with $p : BR \to K(\mathbb{Z}, 3)$ and σ_3 the generator). Proposition L turns it into the bijectivity of r^* into $\Omega K(A, 3) \simeq_* K(A, 2)$.

The lifting lemma. For a source Q, q_Q is a bijection provided (E) for every $F : Q \to_* K(A,2)$, $F;k$ is merely null, and (U) every map $Q \to_* \Omega K(B,4)$ is merely null (`MapFib.Lifting`). For $Q = \Omega S^3$, (U) holds because, by Lemma 1, such a map is the looping of a map $S^3 \to_* K(B,4)$, which is null since $\Omega^3 K(B,4)$ is connected; and (E) follows from (E) at R, because every class from ΩS^3 is $r;c$ with c from R (bijectivity of r^* into $K(A,2)$). For $Q = R$, (U) is $[R, K(B,3)]_* = 0$ (proved for the earlier proof, from (H3) and (H4)). Everything therefore rests on:

(E) at R. *For every $F : R \to_* K(A,2)$, $F;k$ is merely null.*

The fibres of $p_R = \mathsf{fpr} : R \to K(\mathbb{Z},2)$ are 2-connected, so F factors as $p_R;G$ for some $G : K(\mathbb{Z},2) \to_* K(A,2)$. By (H4), $[G;k] = \Phi_B(h) = [\kappa; K(h,4)]$ for a unique $h : \mathbb{Z} \to B$. If $2h(1) = 0$, then $h = \rho;g$ for some $g : \mathbb{Z}/2 \to B$, so $[G;k] = [\bar{\kappa}; K(g,4)]$, and $F;k = p_R;\bar{\kappa};K(g,4)$ is merely null because $p_R;\bar{\kappa}$ is. So (E) at R follows from:

Statement 4 (T)**.** Let (A,B,k) be deloopable, $G : K(\mathbb{Z},2) \to_* K(A,2)$, and $[G;k] = \Phi_B(h)$. Then $2h(1) = 0$.

This is where deloopability is used, and the only place.

5. Statement T

The cross term. For $f : \Omega K(\mathbb{Z},3) \to_* \Omega K(B,5)$, the *cross term* measures the failure of f to be a homomorphism:

$$\Delta_q^f(p) = (fp)^{-1} \cdot f(p \cdot q) \cdot (fq)^{-1}.$$

Packaged through e_2 and e_4, it is a bipointed map $\delta_f : K(\mathbb{Z},2) \to_* \mathrm{Map}_*(K(\mathbb{Z},2), K(B,4))$ (`CrossTerm.CrossGen`). Two facts from the earlier proof are used as black boxes:

(a) if $\delta_f = 0$ then f deloops: $f = \mathsf{loop}\hat{f}$ (the cells argument, `DeloopR.UnloopGen`);
(b) Lemma N: if $f_F = e_2;F;e_4^{-1}$ deloops, where $F = \kappa;K(h,4)$, then $2h(1) = 0$ (`Necessity.Conv`).

So Statement T follows once we show $\delta_{f'} = 0$ for $f' = f_{G;k}$; then f' deloops, $G;k = \kappa;K(h,4)$ merely, and (b) applies. In fact we obtain the stronger conclusion that $f_{G;k}$ deloops.

Detecting the cross term. By the bilinear classification (Theorem B of `doc/h4.pdf`), a bipointed map c is determined by its *evaluation on the generator*: $E(c)(x) \in \Omega^2 K(B,4)$, the looping of c evaluated at the generator of the circle. Here this takes a concrete form: with $\hat{\gamma}_0 \in \Omega^3 K(\mathbb{Z},3)$ the generator and $\lambda_\gamma(m) = (\mathsf{loop}^2 m)(\gamma)$,

$$E(\delta_f)(x) = (\mathsf{loop}^2 e_4)\big(\lambda_{\hat{\gamma}_0}(\Delta_q^f)\big), \qquad q = e_2^{-1}x.$$

So $\delta_f = 0$ as soon as $\lambda_{\hat{\gamma}_0}(\Delta_q^f) = \mathsf{refl}$ for all q. Only the underlying function of Δ_q^f matters: λ is insensitive to the pointing (an Eckmann–Hilton argument at two levels).

The H-space argument. Let $\epsilon : \Omega Y \simeq_* \mathrm{Fib}k$ witness deloopability and

$$\pi = \epsilon;\mathsf{fpr};e_2^{-1} : \Omega Y \to_* \Omega K(A,3), \qquad f_k = e_2;k;e_4^{-1}.$$

By Lemma 1, $\pi = \mathsf{loop}P$ for some $P : Y \to K(A,3)$; hence π is *multiplicative*. This is the one consequence of deloopability we use. Three observations:

(1) $\pi;f_k = \epsilon;(\mathsf{fpr};k);e_4^{-1}$ is merely null, since $\mathsf{fpr};k$ is.
(2) For a multiplicative L, $\Delta_q^{L;f}(p) = \Delta_{Lq}^f(Lp)$. Hence $\Delta_{\pi b}^{f_k} \circ \pi = \Delta_b^{\pi;f_k}$, which is (pointwise) $\Delta_b^0 = \mathsf{refl}$.
(3) $f' = \hat{G};f_k$ with $\hat{G} = e_2;G;e_2^{-1} = \mathsf{loop}K(a,3)$ (since $G = K(a,2)$), again multiplicative.

Choose $\gamma' \in \Omega^3 Y$ with $(\mathsf{loop}^2\pi)\gamma' = (\mathsf{loop}^2\hat{G})\hat{\gamma}_0$; it exists merely because $\mathsf{loop}^2\mathsf{fpr}$ is surjective (the fibre of fpr is 2-connected). Define $\varphi(z) = \lambda_{\gamma'}(\pi; \Delta_z^{f_k})$ for $z \in \Omega K(A,3)$. By (1) and (2), φ vanishes on the image of π, i.e. along fpr; since its target $\Omega^3 K(B,5) \simeq K(B,2)$ is a 2-type and the fibres of fpr are 2-connected, φ vanishes everywhere. Then

$$\lambda_{\hat{\gamma}_0}(\Delta_q^{f'}) = \lambda_{\hat{\gamma}_0}(\hat{G}; \Delta_{\hat{G}q}^{f_k}) = \lambda_{(\mathsf{loop}^2\hat{G})\hat{\gamma}_0}(\Delta_{\hat{G}q}^{f_k}) = \lambda_{\gamma'}(\pi; \Delta_{\hat{G}q}^{f_k}) = \varphi(\hat{G}q) = \mathsf{refl},$$

using (3), naturality of λ, and the choice of γ'. Hence $E(\delta_{f'}) = E(0)$ pointwise and $\delta_{f'} = 0$.

The two mere choices (the null-homotopy in (1) and γ') are eliminated into the goal $\Theta\,\delta_{f'} = \Theta\,0$, an equality in the *set* $\mathrm{Map}_*(K(\mathbb{Z},2), K(B,2))$ (Θ is E followed by $\Omega^2 K(B,4) \simeq K(B,2)$); pointwise equality of E alone would not be a proposition.

Remark 5 (classically). The argument says: the cross term of k, a class in $\tilde{H}^4(K(A,2) \wedge K(A,2); B)$, vanishes after pullback to $\mathrm{Fib}k \wedge \mathrm{Fib}k$ because $\mathrm{Fib}k$ is an H-space over $K(A,2)$ (it is a loop space, and fpr is a loop map up to e_2), and pullback is injective on the $\pi_2 \otimes \pi_2$ part, which is what E detects. The cross term of $\kappa; K(h,4)$ is $2h\,\iota \otimes \iota$; Lemma N is used instead of this computation.

6. Assembly

Putting the pieces together: for every $Y \in \mathsf{SC}_4$, $\mathsf{M}(Y)$ holds (`MarkDirect.Direct.marked`). At $T = \|S^3\|_4$, which is 2-connected and 4-truncated, this is the universal property needed by the Yoneda argument; with $\mathsf{M}(BR)$ and $\pi_4 BR \cong \mathbb{Z}/2$ it gives

$$\pi_4(S^3) \cong \pi_4\|S^3\|_4 \cong \pi_4(BR) \cong \mathbb{Z}/2.$$

7. Relation to the intended route

What is re-proved. The development re-proves Owen Milner's classification equivalence in its simply connected 3-type case,

$$\mathsf{SC}_3 \;\simeq\; \textstyle\sum_{A,B:\mathrm{Ab}} \mathrm{Map}_*\big(K(A,2), K(B,4)\big),$$

following Hart–Milner (Lemma 40, Theorem 41, Note 42 at $n = 2$, over a simply connected base) (`PostnikovClassification.classification`). The argument above *uses* this classification, on objects and equivalences only, together with Wärn's looping embedding (Theorem 2) and the direct 2-torsion argument (Statement T); it does not re-prove the classification a second time.

What is not done. The development does not classify *arbitrary pointed maps* between realised triples, i.e. it does not describe $\mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$ through maps of coefficient groups together with compatible homotopies between the k-invariants, and so it does not obtain the resulting equivalence of homotopy categories.

Consequence. The route Wärn had in mind includes exactly that step: describe arbitrary maps between realised triples in terms of the triple data, then identify the universal object of the sphere on the triple side. The checked proof keeps his classification-and-looping perspective but bypasses the general step: the maps it needs are handled by specialised mapping theorems (the lifting lemma along fpr at the sources R, ΩS^3, and, for $\mathsf{M}(BR)$, BR and S^3; Section 4). It is therefore a different implementation of the idea, not a formalisation of the intended proof.

What remains for the intended formulation. (i) The general correspondence on morphisms: pointed maps $\mathrm{Fib}k \to_* \mathrm{Fib}k'$ (up to homotopy, or as a type) in terms of group maps and compatible homotopies of k-invariants, compatible with identities and composition, so that the classification becomes an equivalence of (homotopy) categories, and its looping counterpart for SC_4. (ii) A reformulation of the universal-property argument through it: the marked property of BR stated and proved on the triple side.

8. What is formalised

Everything above is checked in Unit, development `sag/pi3-s2-postnikov` (`./segc sag/pi3-s2-postnikov all`). The new modules are:

`DeloopEmbedding`	Theorem 2, deloopable triples, $\sum_t \mathsf{Deloopable}(t) \simeq \mathsf{SC}_4$
`MarkLoop`	looping on classes; the class r; Proposition L
`CrossDeloop`	Δ_q^f, λ, pointing irrelevance, the formula for $E(\delta_f)$
`TwoTorsion`	Statement T: $f_{G;k}$ deloops (`TwoTors.unloop`) and $2h(1) = 0$ (`TwoTors.Two.two-tors`); no sphere axiom
`MarkDirect`	(E) at R, the lifting lemma at R and ΩS^3, $\mathsf{M}(Y)$ for every $Y \in \mathsf{SC}_4$
`Pi4Direct`	the assembly, `Pi4S3.pi4-S3 : GIso[pi4 S3 base, ab-grp Z2ab]`

The final theorem depends only on the universal property of S^3 (a module parameter) and the Theory parameters (univalence, replacement). It does not use the Postnikov presentation of $\|S^3\|_4$ (`Presentation4`): a copy of the development with that module deleted still checks. Detailed informal proofs, declaration by declaration, are in `doc/delooping-direct.pdf`; the earlier route is `doc/pi4-s3.pdf`.

References

[1] D. Wärn. *Eilenberg–MacLane spaces and stabilisation in homotopy type theory.* arXiv:2301.03685, 2023.
[2] P. Hart and O. Milner. *Classifying 2-groups in homotopy type theory.* LICS 2026, LIPIcs vol. 380.